Experiment-15

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
data = {
   "Name": ["Arun", "Bala", "Charan", "Divya", "Ezhil", "Bala", "Farah", "Gokul"],
"Age": [20, 21, np.nan, 22, -5, 21, 20, 23],
   "Department": ["CSE", "cse", "ECE", "ECE", "CSE", "cse", "IT", "IT"],
"Marks": [85, 90, 75, np.nan, 120, 90, 65, 80]
}
df = pd.DataFrame(data)
print(df)
```

```
     Name   Age Department  Marks
0    Arun  20.0        CSE   85.0
1    Bala  21.0        cse   90.0
2  Charan   NaN        ECE   75.0
3   Divya  22.0        ECE    NaN
4   Ezhil  -5.0        CSE  120.0
5    Bala  21.0        cse   90.0
6   Farah  20.0         IT   65.0
7   Gokul  23.0         IT   80.0
```

```
print("Missing Values:")
print(df.isnull().sum())
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Marks"] = df["Marks"].fillna(df["Marks"].median())
print("\nAfter Handling Missing Values:")
print(df)
```

```
Missing Values:
Name          0
Age           1
Department    0
Marks         1
dtype: int64

After Handling Missing Values:
     Name   Age Department  Marks
0    Arun  20.0        CSE   85.0
1    Bala  21.0        cse   90.0
2  Charan  21.0        ECE   75.0
3   Divya  22.0        ECE   85.0
4   Ezhil  -5.0        CSE  120.0
5    Bala  21.0        cse   90.0
6   Farah  20.0         IT   65.0
7   Gokul  23.0         IT   80.0
```

```
df.loc[df["Age"] < 0, "Age"] = np.nan

df.loc[(df["Marks"] < 0) | (df["Marks"] > 100), "Marks"] = np.nan

df["Age"] = df["Age"].fillna(df["Age"].median())

df["Marks"] = df["Marks"].fillna(df["Marks"].median())

print(df)
```

```
     Name   Age Department  Marks
0    Arun  20.0        CSE   85.0
1    Bala  21.0        cse   90.0
2  Charan  21.0        ECE   75.0
3   Divya  22.0        ECE   85.0
4   Ezhil  21.0        CSE   85.0
5    Bala  21.0        cse   90.0
6   Farah  20.0         IT   65.0
7   Gokul  23.0         IT   80.0
```

```
df = df.drop_duplicates()

print("Dataset After Removing Duplicates:")

print(df)
```

```
Dataset After Removing Duplicates:
     Name   Age Department  Marks
0    Arun  20.0        CSE   85.0
1    Bala  21.0        cse   90.0
2  Charan  21.0        ECE   75.0
3   Divya  22.0        ECE   85.0
4   Ezhil  21.0        CSE   85.0
6   Farah  20.0         IT   65.0
7   Gokul  23.0         IT   80.0
```

```
df["Department"] = (df["Department"].str.strip().str.upper())

print(df["Department"])
```

```
0    CSE
1    CSE
2    ECE
3    ECE
4    CSE
6     IT
7     IT
Name: Department, dtype: object
```